Arch Linux VM Installation Documentation

Overview

For this assignment, I manually installed Arch Linux in a virtual machine using the official Arch Linux Installation Wiki and google when raging.

My completed VM uses:

- Arch Linux
- Xfce desktop environment
- SDDM display manager
- Bash
- OpenSSH
- Automatic graphical boot
- Terminal color coding
- Custom Bash aliases
- Additional Xfce utilities and themes

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1. Verify the Installation Environment

```
ls /sys/firmware/efi/efivars
```

Checks that the VM booted using UEFI.

```
ping -c 4 archlinux.org
```

Checks network connectivity.

```
timedatectl
```

Checks system time and synchronization.

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2. Identify and Partition the Disk

```
lsblk
```

Displays disks and partitions.

```
cfdisk /dev/sda
```

Used to create the EFI System Partition and Linux root partition.

> The exact disk name may be different depending on the VM configuration.

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3. Format the Partitions

```
mkfs.fat -F 32 /dev/sda1
```

Formats the EFI partition as FAT32.

```
mkfs.ext4 /dev/sda2
```

Formats the root partition as ext4.

---

4. Mount the Filesystems

```
mount /dev/sda2 /mnt
```

Mounts the root filesystem.

```
mkdir -p /mnt/boot
```

Creates the boot directory.

```
mount /dev/sda1 /mnt/boot
```

Mounts the EFI partition.

```
lsblk
```

Verifies the partitions and mount points.

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5. Install Arch Linux

```
pacstrap -K /mnt base linux linux-firmware
```

Installs the base Arch Linux system, kernel, and firmware.

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6. Generate the Filesystem Table

```
genfstab -U /mnt >> /mnt/etc/fstab
```

Creates the filesystem configuration used during boot.

```
cat /mnt/etc/fstab
```

Checks the generated configuration.

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7. Enter the New Arch System

```
arch-chroot /mnt
```

Changes into the newly installed Arch system.

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8. Configure Time and Locale

```
ln -sf /usr/share/zoneinfo/America/Chicago /etc/localtime
```

Sets the time zone.

```
hwclock --systohc
```

Synchronizes the hardware clock.

```
nano /etc/locale.gen
```

I enabled:

```
en_US.UTF-8 UTF-8
```

Then ran:

```
locale-gen
```

Created the locale configuration:

```
echo "LANG=en_US.UTF-8" > /etc/locale.conf
```

---

9. Configure the Hostname

My hostname is:

```
archlinux
```

The hostname is configured with:

```
echo "archlinux" > /etc/hostname
```

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10. Set the Root Password

```
passwd
```

Sets the password for the root account.

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11. Install GRUB

```
pacman -S grub efibootmgr
```

Installs GRUB and EFI boot management tools.

```
grub-install --target=x86_64-efi --efi-directory=/boot --bootloader-id=GRUB
```

Installs GRUB for UEFI.

```
grub-mkconfig -o /boot/grub/grub.cfg
```

Generates the GRUB configuration.

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12. Create User Accounts

My normal user account is:

```
mod
```

I also created the required `codi` account.

```
useradd -m -G wheel mod
passwd mod
```

```
useradd -m -G wheel codi
passwd codi
```

The `wheel` group is used for sudo access.

---

13. Configure sudo

```
pacman -S sudo
```

Installs sudo.

```
EDITOR=nano visudo
```

I enabled:

```
%wheel ALL=(ALL:ALL) ALL
```

This allows users in the `wheel` group to use `sudo`.

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14. Finish the Base Installation

```
exit
```

Exits the chroot environment.

```
umount -R /mnt
```

Unmounts the installed filesystems.

```
reboot
```

Reboots into the installed Arch system.

Post-Installation Configuration

15. Update Arch Linux

```
sudo pacman -Syu
```

Updates installed packages and package databases.

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16. Install Xfce

```
sudo pacman -S xorg-server xfce4 xfce4-goodies
```

Installs:

- Xorg graphical display system
- Xfce desktop environment
- Additional Xfce utilities and plugins

I chose **Xfce** because it is lightweight while still providing a complete desktop environment.

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17. Install SDDM

```
sudo pacman -S sddm
```

Installs the SDDM graphical login manager.

```
sudo systemctl enable sddm.service
```

Configures SDDM to start automatically.

I also created an SDDM configuration directory:

```
sudo mkdir -p /etc/sddm.conf.d
```

Then edited the Xfce configuration:

```
sudo nano /etc/sddm.conf.d/10-xfce.conf
```

---

18. Boot Into the GUI Automatically

```
sudo systemctl set-default graphical.target
```

Sets the graphical environment as the default boot target.

I verified it with:

```
systemctl get-default
```

I also checked SDDM:

```
systemctl is-enabled sddm.service
```

Then rebooted:

```
sudo reboot
```

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19. Install and Enable SSH

```
sudo systemctl enable --now sshd
```

Enables the SSH server at boot and starts it immediately.

I checked the VM's IPv4 address with:

```
ip addr
```

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20. Configure Bash

I checked my current shell with:

```
echo $SHELL
```

I also checked whether Bash was installed with:

```
command -v bash
```

My shell configuration is stored in:

```
~/.bashrc
```

I edited it using:

```
sudo nano ~/.bashrc
```

After making changes, I reloaded it with:

```
source ~/.bashrc
```

---

21. Add Terminal Color Coding

I configured Bash so commands such as `ls` display color output.

An example command I used while testing this was:

```
ls --color=auto /usr/bin | head -30
```

I also added color directly to my `ll` alias.

---

22. Add Bash Aliases

I added the following aliases to `~/.bashrc`:

```
# Custom Arch Linux aliases
alias update='sudo pacman -Syu'
alias ll='ls -lah --color=auto'
alias ..='cd ..'
alias ...='cd ../..'
alias c='clear'
alias myip='ip -br addr'
```

I then applied the changes:

```
source ~/.bashrc
```

I checked the configured aliases with:

```
alias
```

Alias Purposes

Alias	Command	Purpose
`update`	`sudo pacman -Syu`	Updates Arch Linux
`ll`	`ls -lah --color=auto`	Detailed colored file listing
`..`	`cd ..`	Move up one directory
`...`	`cd ../..`	Move up two directories
`c`	`clear`	Clears the terminal
`myip`	`ip -br addr`	Displays IP addresses in a short format

23. VM Integration

I installed the SPICE guest agent:

```
sudo pacman -S spice-vdagent
```

This improves integration between the virtual machine and the host.

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24. Customize Xfce

I attempted to install additional themes:

```
sudo pacman -S arc-gtk-theme papirus-icon-theme
```

I later tried:

```
sudo pacman -S materia-gtk-theme papirus-icon-theme
```

I also installed the Xfce Whisker Menu plugin:

```
sudo pacman -S xfce4-whiskermenu-plugin
```

---

25. Install Additional Utilities

```
sudo pacman -S fastfetch btop htop git vim
```

Installed several system and administration tools.

I tested Fastfetch with:

```
fastfetch
```

I also installed additional Xfce monitoring tools:

```
sudo pacman -S xfce4-systemload-plugin xfce4-taskmanager
```

Problems and Recovery

GUI Did Not Start Automatically

After installing Xfce and SDDM, I needed to make sure the system booted directly into the graphical environment.

I used:

```
sudo systemctl enable sddm.service
sudo systemctl set-default graphical.target
```

Then rebooted:

```
sudo reboot
```

SDDM Configuration

I needed to create an SDDM configuration directory manually:

```
sudo mkdir -p /etc/sddm.conf.d
```

I then edited:

```
sudo nano /etc/sddm.conf.d/10-xfce.conf
```

I rebooted several times while testing the graphical login configuration.

Theme Installation

I attempted to install:

```
sudo pacman -S arc-gtk-theme papirus-icon-theme
```

and later:

```
sudo pacman -S materia-gtk-theme papirus-icon-theme
```

This was part of customizing the Xfce desktop and testing which theme packages were available.
